

DBMS Notes: Types of Attributes in ER Model

Comprehensive Classification • ER Symbols • GATE & University Exam Guide

HIGH FREQUENCY EXAM TOPIC ★★★

Attributes are properties or characteristics that describe an entity. For example, for a **STUDENT** entity, attributes can include Roll_No, Name, Age, Address, Mobile_No, and Date_of_Birth.

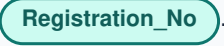
🧠 ER Attribute Symbols — Must Memorize ★★★

Attribute Category	Description	ER Visual Symbol	Symbol Representation
Simple / Single-Valued	Standard property with one value	Single Oval	
Multi-Valued	Can hold multiple values	Double Oval	
Composite	Divided into sub-components	Oval linked to Sub-Ovals	 → (First, Last)
Derived	Calculated from another stored value	Dashed / Dotted Oval	
Key Attribute	Uniquely identifies each entity instance	Underlined Text inside Oval	
Non-Key Attribute	Does not guarantee unique identification	Normal Oval	

1. Single-Valued vs. Multi-Valued Attributes ★★★

A. Single-Valued Attribute

Can have only **one single value** for a particular entity instance at any given time.

- **Examples:** Registration_No, Roll_No, Date_of_Birth, Gender.
- **ER Symbol:** Single Oval .

B. Multi-Valued Attribute

Can hold **multiple values** for a single entity instance.

- **Examples:** Mobile_No (a student can have 2 phone numbers), Email_ID, Address (Permanent & Correspondence).
- **ER Symbol:** Double Oval.

2. Simple vs. Composite Attributes ★★★

A. Simple (Atomic) Attribute

An attribute that **cannot be divided** into smaller meaningful sub-components.

- **Examples:** Age, Gender, Roll_No.

B. Composite Attribute

An attribute that **can be divided** into smaller meaningful sub-attributes.

- **Examples:**
 - Name → First_Name, Middle_Name, Last_Name
 - Address → House_No, Street, City, State, PIN_Code

3. Stored vs. Derived Attributes ★★★

A. Stored Attribute

An attribute whose value is directly supplied and stored permanently in the database storage.

- **Example:** Date_of_Birth (e.g., 15-08-2006).

B. Derived Attribute

An attribute whose value is dynamically calculated or derived from another stored attribute or system parameter.

Real-World Logic

Formula: Stored Attribute (Date_of_Birth) + System Date → Calculated Value (Age). We don't need to store Age manually because it changes automatically over time.

- **ER Symbol:** Dashed / Dotted Oval .

4. Key vs. Non-Key Attributes ★★★

A. Key Attribute

An attribute that **uniquely identifies** a specific entity instance from an entity set. No two entities can share the same key attribute value.

- **Examples:** Registration_No, SSN, Employee_ID.
- **ER Symbol:** Text inside Oval is Underlined .

B. Non-Key Attribute

An attribute that does not uniquely identify an entity and may contain repeated/duplicate values across rows.

- **Examples:** Name, Age, City, Gender.

5. Additional Attribute Classifications

A. Required vs. Optional Attributes

- **Required Attribute:** Must contain a value (Cannot be NULL). Enforced in SQL as NOT NULL (e.g., Student_Name).
- **Optional Attribute:** Can be left blank or omitted if data is unavailable (e.g., Middle_Name).

B. Complex Attribute ★★★

An attribute that is both **Composite** (can be broken into sub-parts) and **Multi-Valued** (can have multiple sets of values).

Complex Attribute Formula

Complex Attribute = Composite + Multi-Valued

Example: A student having multiple addresses where each address consists of (House_No, Street, City, State).

6. Complete Attribute Classification Matrix

Comprehensive breakdown of attributes for a single **STUDENT** entity:

Attribute Name	Attribute Classification Type	Reasoning / Justification
Registration_No	Key Attribute	Uniquely identifies each student entity instance (Underlined).
Name	Composite Attribute	Can be divided into First_Name, Middle_Name, Last_Name.
Date_of_Birth	Stored Attribute	Directly stored value in database.
Age	Derived Attribute	Dynamically computed from Date_of_Birth and current system date (Dashed Oval).
Mobile_No	Multi-Valued Attribute	A student can possess more than one phone number (Double Oval).
Gender	Simple / Single-Valued	Atomic, cannot be divided, and holds a single value.
Middle_Name	Optional Attribute	May be left NULL or omitted.
Addresses	Complex Attribute	Composite (House, Street, City) + Multi-valued (Multiple locations).



Exam & Interview Rapid-Fire Questions

Q1. Which attribute is represented by a double oval in an ER diagram?

→ **Multi-Valued Attribute.**

Q2. Which attribute is represented by a dashed/dotted oval?

→ **Derived Attribute.**

Q3. How is a Key attribute distinguished in ER diagrams?

→ **By underlining the text inside the oval.**

Q4. A student's age calculated from Date of Birth is classified as what?

→ **Derived Attribute.**

Q5. An attribute consisting of First Name, Middle Name, and Last Name is?

→ **Composite Attribute.**

Q6. What is an attribute called if it is both Composite and Multi-Valued?

→ **Complex Attribute.**

★ Quick Revision Cheat Sheet

- **Single-valued:** 1 value | **Multi-valued:** Multiple values (Double Oval)
- **Simple:** Cannot divide | **Composite:** Can divide into sub-parts
- **Stored:** Directly saved | **Derived:** Computed (Dashed Oval)
- **Key:** Unique identification (Underlined) | **Non-key:** Can duplicate
- **Complex:** Composite + Multi-valued combination